

# CS & IT ENGINEERING



## Data Structure & Programming

### Hash Tables

Lecture No.- 02



By- Satya sir

# Recap of Previous Lecture



- Hashing

- Hash Function

- Collision

- Collision Resolution Techniques

- Open Hashing / Separate chaining Method.

- Closed Hashing / Open addressing

[ Linear Probing  $h(k, i) = (k + i) \cdot / n$  : Primary clustering  
Quadratic Probing  
Double Hashing ]



# Topics to be Covered



- Quadratic Probing
  - Hash Function
  - Drawback
- Double Hashing
  - Hash Function
  - Comparison of Techniques
- PYQ Practice





## Topic : Hash Tables - 2



#Q. Table size = 13 Hash Function,  $h(k, i) = (3k + i) \cdot n$

Keys: 71, 92, 64, 18, 5, 11, 17, 23, 45

Linear Probing scheme to be Used for Collision Resolution.

$$i=0 \Rightarrow h(k, i) = (3k + 0) \cdot n = 3k \cdot n$$

$$h(71, 0) = 213 \cdot n = 5$$

$$h(92, 0) = 276 \cdot n = 3$$

$$h(64, 0) = 192 \cdot n = 10$$

$$h(18, 0) = 54 \cdot n = 2$$

$$h(5, 0) = 15 \cdot n = 2 \text{ collision}$$

$$h(5, 1) = (3(5) + 1) \cdot n = 3 \text{ collision}$$

$$h(5, 2) = (3(5) + 2) \cdot n = 4$$

$$h(11, 0) = 33 \cdot n = 7$$

$$h(17, 0) = 51 \cdot n = 12$$

$$h(23, 0) = 69 \cdot n = 4 \text{ collision}$$

$$h(23, 1) = 70 \cdot n = 5 \text{ collision}$$

$$h(23, 2) = 71 \cdot n = 6$$

$$h(45, 0) = 135 \cdot n = 5 \text{ collision}$$

$$h(45, 1) = 136 \cdot n = 6 \text{ collision}$$

$$h(45, 2) = 137 \cdot n = 7 \text{ collision}$$

$$h(45, 3) = 138 \cdot n = 8$$

|    |    |
|----|----|
|    | 0  |
|    | 1  |
| 18 | 2  |
| 92 | 3  |
| 5  | 4  |
| 71 | 5  |
| 23 | 6  |
| 11 | 7  |
| 45 | 8  |
|    | 9  |
| 64 | 10 |
|    | 11 |
| 17 | 12 |





Consider a hash table of size 11 that uses open addressing with linear probing. Let  $h(k) = k \bmod 11$  be the hash function used. A sequence of records with keys

43 36 92 87 11 4 71 13 14

is inserted into an initially empty hash table, the bins of which are indexed from zero to ten. What is the index of the bin into which the last record is inserted?

- (A) 2
- (B) 4
- (C) 6
- (D) 7

$$h(k) = k \cdot 11$$

$$h(43) = 43 \cdot 11 = 10$$

$$h(36) = 3$$

$$h(92) = 4$$

$$h(87) = 10 \text{ collision} \rightarrow 0$$

$$h(11) = 0 \text{ collision} \rightarrow 1$$

$$h(4) = 4 \text{ collision} \rightarrow 5$$

$$h(71) = 5 \text{ collision} \rightarrow 6$$

$$h(13) = 2$$

$$h(14) = 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow \underline{7}$$

|    |    |
|----|----|
| 0  | 87 |
| 1  | 11 |
| 2  | 13 |
| 3  | 36 |
| 4  | 92 |
| 5  | 4  |
| 6  | 71 |
| 7  | 14 |
| 8  |    |
| 9  |    |
| 10 | 43 |



2      4      1      9      9      1      3      9

Given the following input (4322, 1334, 1471, 9679, 1989, 6171, 6173, 4199) and the hash function  $x \bmod 10$ , which of the following statements are true?

⇒ Chaining Method is Used

1. 9679, 1989, 4199 hash to the same value ✓
2. 1471, 6171 hash to the same value ✓
3. All elements hash to the same value ✗
4. Each element hashes to a different value ✗

- (A) 1 only  
(B) 2 only  
(C) 1 and 2 only ✓  
(D) 3 or 4





Consider a hash table of size seven, with starting index zero, and a hash function  $(3x + 4) \bmod 7$ . Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that '\_' denotes an empty location in the table. (Linear Probing)

- (A) 8, \_, \_, \_, \_, \_ 10
- ☒ (B) 1, 8, 10, \_, \_, \_ 3
- (C) 1, \_, \_, \_, \_, \_ 3
- (D) 1, 10, 8, \_, \_, \_ 3

| 0 | 1 | 2  | 3 | 4 | 5 | 6 |
|---|---|----|---|---|---|---|
| 1 | 8 | 10 | - | - | - | 3 |

$$h(1) = (3(1) + 4) \cdot 7 = 7 \cdot 7 = 0$$

$$h(3) = 13 \cdot 7 = 6$$

$$h(8) = 28 \cdot 7 = 0 \text{ collision} \rightarrow 1$$

$$h(10) = 34 \cdot 7 = 6 \rightarrow 0 \rightarrow 1 \rightarrow 2$$





Which one of the following hash functions on integers will distribute keys most uniformly over 10 buckets numbered 0 to 9 for  $i$  ranging from 0 to 2020?

- ~~(A)~~  $h(i) = i^2 \bmod 10$   
(B)  $h(i) = i^3 \bmod 10$  ✓  
~~(C)~~  $h(i) = (11 * i^2) \bmod 10$   
~~(D)~~  $h(i) = (12 * i) \bmod 10$

LSB digit: 0 to 9 only

| $i$<br>LSB digit | $i^2 \div 10$ | $i^3 \div 10$ | $11 * i^2$ | $12 * i$ |
|------------------|---------------|---------------|------------|----------|
| 0                | 0             | 0             | 0          | 0        |
| 1                | 1             | 1             | 1          | 2        |
| 2                | 4             | 8             | 4          | 4        |
| 3                | 9             | 7             | 9          | 6        |
| 4                | 6             | 4             | 6          | 8        |
| 5                | 5             | 5             | 5          | 0        |
| 6                | 6             | 6             | 6          | 2        |
| 7                | 9             | 3             | 9          | 4        |
| 8                | 4             | 2             | 4          | 6        |
| 9                | 1             | 9             | 1          | 8        |

Slot 1, 3, 5, 7, 9 are Not mapped to any key

2, 3, 7, 8      2, 3, 7, 8

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## Topic : Hash Tables - 2



$$h'(k) = k \cdot m$$

### Quadratic Probing

Drawback: Secondary clustering

⇒ Non consecutive slots form a group and increase probes for keys

$$h(k, i) = (h'(k) + c_1 i^1 + c_2 i^2) \bmod m$$

$i=0$  initially for each key. If collision for a key, then  $i=i+1$

when  $i=0$

$$h(k, 0) = (h'(k)) \cdot m = (k \cdot m) \cdot m = \underline{k \cdot m}$$

Ex: Consider inserting the keys 74, 28, 36, 58, 21, 64 into a hash table of size  $m=11$  using quadratic probing with  $c_1=1$  and  $c_2=3$ . Further consider that the primary hash function is  $h'(k) = k \bmod m$ .

$$h(74, 0) = (74 \cdot 11) \cdot 11 = 8 \cdot 11 = 8$$

$$h(28, 0) = 28 \cdot 11 = 6$$

$$h(36, 0) = 36 \cdot 11 = 3$$

$$h(58, 0) = 58 \cdot 11 = 3 \text{ collision}$$

$$h(58, 1) = [(58 \cdot 11) + (1 \cdot 1) + (3 \cdot 1^2)] \cdot 11 = (3 + 1 + 3) \cdot 11 = 7$$

$$h(21, 0) = 21 \cdot 11 = 10$$

$$h(64, 0) = 64 \cdot 11 = 9$$

$$h(47, 0) = 47 \cdot 11 = 3 \text{ collision}$$

$$h(47, 1) = (3 + 1 + 3) \cdot 11 = 7 \text{ collision}$$

$$h(47, 2) = (3 + 1 \cdot 2 + 3 \cdot 2^2) \cdot 11 = 17 \cdot 11 = 6 \text{ coll}$$

$$h(47, 3) = (3 + 1 \cdot 3 + 3 \cdot 3^2) \cdot 11 = 33 \cdot 11 = 0$$

$$h(80, 0) = 3 \text{ collision}$$

$$h(80, 1) = 7 \text{ collision}$$

$$h(80, 2) = 6 \text{ collision}$$

$$h(80, 3) = 0 \text{ coll}$$

$$h(80, 4) = (3 + 4 + 48) \cdot 11$$

$$= 55 \cdot 11 = 0 \text{ coll}$$

$$h(80, 5) = (3 + 5 + 75) \cdot 11$$

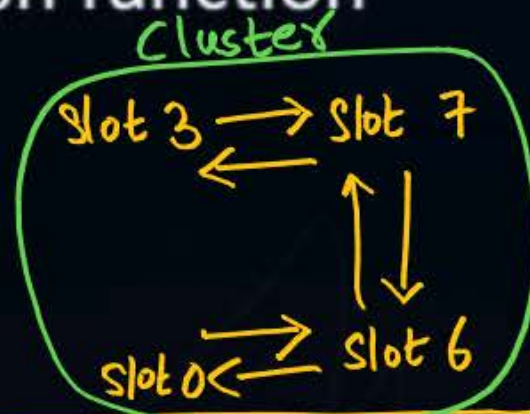
$$= 83 \cdot 11 = 6 \text{ coll}$$

$$h(80, 6) = (3 + 6 + 108) \cdot 11 = 117 \cdot 11 = 7 \text{ collision}$$

$$h(80, 7) = (3 + 7 + 147) \cdot 11 = 157 \cdot 11 = 3 \text{ collision}$$

$$h(80, 8) = 5$$

|    |    |
|----|----|
| 0  | 47 |
| 1  |    |
| 2  | 14 |
| 3  | 36 |
| 4  |    |
| 5  | 80 |
| 6  | 28 |
| 7  | 58 |
| 8  | 74 |
| 9  | 64 |
| 10 | 21 |



$$h(14, 0) = 3$$

$$h(14, 1) = 7$$

$$h(14, 2) = 6$$

$$h(14, 3) = 0$$

$$h(14, 4) = 0$$

$$h(14, 5) = 6$$

$$h(14, 6) = 7$$

$$h(14, 7) = 3$$

$$h(14, 8) = 5$$

$$h(14, 9) = 255 \cdot 11 = 2$$





## Topic : Hash Tables - 2



Double hashing uses a hash function of the form

$$h(k, i) = (h_1(k) + i h_2(k)) \bmod m$$

$\Rightarrow$  It is free from clustering issue.

Where  $h_1$  and  $h_2$  are auxiliary hash functions and  $m$  is the size of the hash table.

Primary Hash Function

secondary Hash Function

$$h_1(k) = k \bmod m \text{ or } h_2(k) = k \bmod m'$$

Here  $m'$  is slightly less than  $m$  (say  $m-1$  or  $m-2$ ).

$m' < m$   
(Nearest Prime Number)

Ex:  $m=12, m'=11$

$m=15, m'=13$





Consider a double hashing scheme in which the primary hash function is  $h_1(k) = k \bmod 21$ , and the secondary hash function is  $h_2(k) = 1 + (k \bmod 19)$ . Assume that the table size is 21. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value  $k = 70$  is 0.

$$h(k, i) = [h_1(k) + i * h_2(k)] \cdot / \cdot m$$

$$\begin{aligned} h(70, 1) &= [h_1(70) + 1 * h_2(70)] \cdot / \cdot 21 \\ &= [(70 \cdot / \cdot 21) + 1 * [1 + (70 \cdot / \cdot 19)]] \cdot / \cdot 21 \\ &= (7 + 1 * [1 + 13]) \cdot / \cdot 21 \\ &= 21 \cdot / \cdot 21 = 0 \end{aligned}$$





## Topic : Hash Tables - 2



4/19

74, 28, 36, 58, 21, 64, 47, 80, 14

Double Hashing, Table size = 11

$$h_1(k) = k \% 11 \quad h_2(k) = k \% 7$$





## 2 mins Summary



- Quadratic Probing
- Double Hashing
- PYQ Practice

Stay Tuned ... 

Practice Session on Monday. Prepare Well.



**THANK - YOU**